Averaged Jacobian Regularity Can Misrank Few-Step Flow-Matching Schedules: A Certified Gaussian Counterexample

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Abstract

Few-step sampling in flow matching and stochastic interpolants requires choosing an interpolation schedule before endpoint error is observed. Averaged squared Lipschitzness of the drift, measured by the time-integrated squared 2-norm of its spatial Jacobian, is a plausible schedule-design criterion, but it is not a solver-specific discretization-error functional. Chen, Vanden-Eijnden, and Xu propose minimizing this quantity, denoted A_2; they do not claim or prove that A_2 universally ranks equal-NFE solver error. We study that narrower surrogate question. For independent $N(0,1)$ and $N(0,4)$, the exact regularity integrals are $5\pi/8-1$ and $\pi^2/16$, so regularity prefers trigonometric VP, whereas explicit Heun, a two-stage Runge–Kutta method, with a budget of eight function evaluations (NFE 8), prefers the linear path in Gaussian Wasserstein-2 distance. The inversion is certified by an exact rational Heun product and a nonnegative element of $\mathbb{Q}[\pi,\sqrt{2}]$. A complementary construction shows that, for every step count N and every admissible endpoint log-scale, the unique integrated-regularity minimizer can have strictly larger N-step Euler endpoint error than a higher-regularity competitor aligned with that solver grid. A specified finite Gaussian enumeration of four candidate paths on 36 tested blocks contains both agreement and pairwise disagreement. These results limit a universal surrogate interpretation; they do not evaluate learned velocity fields or estimate a population failure frequency.

1 Introduction

Few-step sampling commits to a path, a solver, and a small NFE budget before any error is observed. Lipschitz constants enter classical one-step bounds [Hairer et al., 1993]. Chen et al. [2025] propose minimizing A_2, the averaged squared Lipschitzness of the drift (their Definition 3.2), as a schedule-design objective in a specified admissible family. They do not claim or prove that A_2 universally ranks equal-NFE solver error. We study that narrower surrogate question on a flow-matching marginal ODE.

The object is an integrated unsigned Jacobian energy. For a fixed finite-step solver, endpoint error depends on stagewise field values, temporal derivatives, quadrature structure, and signed cancellations that this energy does not determine. For independent $N(0,1)$ and $N(0,4)$, exact regularity prefers trigonometric VP while Heun at NFE 8 prefers the linear path in Gaussian W_2 (Section 3). A solver-grid-aware construction then shows that even the global $\mathcal{R}$-minimizer need not minimize fixed-grid Euler error (Section 6); the competitor is built from the solver resolution. A finite Gaussian enumeration of four specified candidate paths on 36 tested blocks contains both agreement and pairwise disagreement (Section 5). No learned field is used.

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Venue relevance is through distribution learning, Wasserstein endpoint error, and the geometry of Gaussian covariance interpolation, not through a new geometric architecture or a practical OT algorithm. Tao and Choi [2026] give a broader dynamical error analysis, including material derivatives, strain, vorticity, logarithmic norms, and learned models. The present contribution is narrower but exact and auditable: a certified scalar Gaussian ranking inversion, a closed-form W_2 comparison, a grid-aware impossibility construction, and a transparent finite enumeration. Contemporaneous Lipschitz Euler rates [Stéphanovitch, 2026] bound a different object.

2 Gaussian interpolants and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent. A shared schedule is a scalar pair (α, σ). We distinguish three regularity classes.

- Class S: $\alpha, \sigma \in C^1([0,1])$ with $\alpha(0) = \sigma(1) = 1$, $\alpha(1) = \sigma(0) = 0$, and $(\alpha(t), \sigma(t)) \neq (0,0)$ on $[0,1]$. Linear and trigonometric VP belong to Class S.
- Class Q: a C^1 path of SPD covariances $Q(t)$ with continuous $A = \frac{1}{2}Q'Q^{-1}$. The per-mode schedule $q_i(t) = \lambda_i^t$ is Class Q.
- Class R: $\alpha^2, \sigma^2 \in C^1([0,1])$, $\alpha^2 \geq 0$, $\sigma^2 \geq 0$, $\alpha^2 + \sigma^2 > 0$, and $\alpha(0)^2 = 1$, $\sigma(0)^2 = 0$, $\alpha(1)^2 = 0$, $\sigma(1)^2 = 1$. Square roots need not be C^1 at the endpoints. Chen Ex. 3.3, $\sigma(t)^2 = (M^t - 1)/(M-1)$ and $\alpha(t)^2 = (M - M^t)/(M-1)$ for $M > 0$, $M \neq 1$, is Class R and Class Q, not Class S. The $M = 1$ limit is $\sigma(t)^2 = t$, $\alpha(t)^2 = 1 - t$.

The interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023] is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$. The marginal field is affine,

$$b(t,x) = A(t)\,x + c(t), \qquad A = C_t Q^{-1}, \qquad c = \dot{m} - Am. \tag{1}$$

The formula $C_t = \frac{1}{2}(\alpha^2)'\,\Sigma_0 + \frac{1}{2}(\sigma^2)'\,\Sigma_1$ extends continuously to $[0,1]$ for Class R; equivalently $C_t = \dot\alpha\alpha\,\Sigma_0 + \dot\sigma\sigma\,\Sigma_1$ on $(0,1)$ where the coefficients are differentiable and nonzero. This is a flow-matching marginal ODE, not a score-based probability-flow ODE in the sense of Song et al. [2021]. If $\Sigma_0 = I$, then Q, Q', A, and Σ_1 share the eigenbasis of Σ_1, and mode i obeys $x_i' = a_i(t)x_i + c_i(t)$ with $q_i = \alpha^2 + \lambda_i\sigma^2$ and $a_i = q_i'/(2q_i)$. The covariance drift formula is independent of the mean path $c(t)$. We compare the linear path $(1-t, t)$ and trigonometric VP $(\cos(\pi t/2), \sin(\pi t/2))$. Following Chen et al. [2025, Definition 3.2], $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2\,dt$. For state-independent A this coincides with their path expectation. By definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2 of the endpoint laws [Gelbrich, 1990]. This is neither pathwise trajectory error nor local truncation error. Equal NFE uses $S = B/s$ steps with $s = 1, 2, 4$ for Euler, Heun, and RK4.

Any regular variance path from 1 to $\lambda > 0$ obeys $\int_0^1 a\,dt = \frac{1}{2}\ln\lambda$. Hence $\mathcal{R} = \int_0^1 a^2\,dt = (\int a)^2 + \mathrm{Var}_t(a) \geq (\frac{1}{2}\ln\lambda)^2$, with equality if and only if a is constant. The one-dimensional minimizer is $q(t) = \lambda^t$, $a \equiv \frac{1}{2}\ln\lambda$, which is Chen et al. Example 3.3. Linear and trigonometric VP are C^1 shared interpolants. In several dimensions, the per-mode schedule $q_i(t) = \lambda_i^t$ yields $\mathcal{R} = (\frac{1}{2}\max_i|\ln\lambda_i|)^2$. When Σ_1 has at least two distinct eigenvalues, that path is not a Class S shared interpolant.

3 A one-dimensional Gaussian ranking counterexample

Assumption 1. *Endpoints $N(0,1)$ and $N(0,4)$ are independent. Schedules are linear and trigonometric VP. The solver is explicit Heun with NFE 8 (four steps of size $1/4$).*

Proposition 1. *Under the assumption, $\mathcal{R}_{\mathrm{lin}} = 5\pi/8 - 1 > \mathcal{R}_{\mathrm{VP}} = \pi^2/16$, while the Heun factors satisfy $r_{\mathrm{lin}} = 6797469/3559400$ and $0 < r_{\mathrm{VP}} < 187/100$. Hence $W_2^{\mathrm{lin}} < 0.091 < 0.130 < W_2^{\mathrm{VP}}$.*

Proof sketch. The linear variance is $q_{\mathrm{lin}}(t) = 1 - 2t + 5t^2$. Completing the square gives the equivalent identities $q_{\mathrm{lin}}(t) = 5\big[(t - 1/5)^2 + (2/5)^2\big] = 5(t - 1/5)^2 + 4/5$. The substitution $u = t - 1/5 = (2/5)\tan\theta$ converts $\int a_{\mathrm{lin}}^2$ into $(5/2)\int \sin^2\theta\,d\theta$. Limits $\arctan(-1/2)$ and $\arctan 2$ together with $\arctan 2 + \arctan(1/2) = \pi/2$ yield $\mathcal{R}_{\mathrm{lin}} = 5\pi/8 - 1$. The VP integral evaluates to $\pi^2/16$ by a Weierstrass identity. The comparison $5\pi/8 - 1 > \pi^2/16$ rearranges to $(\pi-2)(\pi-8) < 0$,

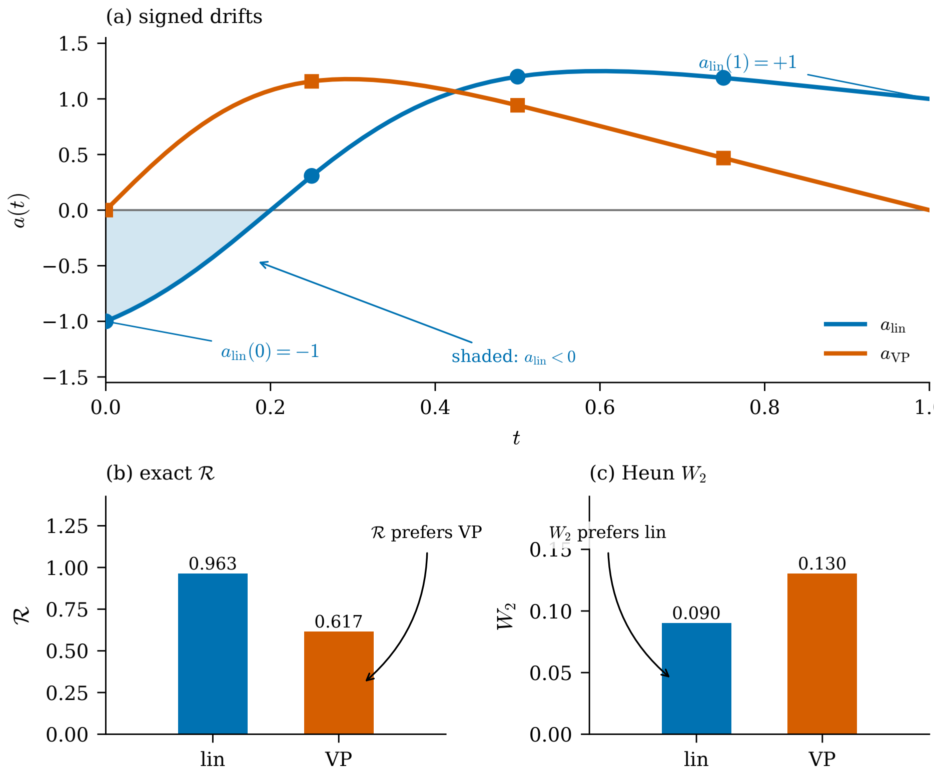

Figure 1: Certified scalar counterexample for independent $N(0, 1)$ and $N(0, 4)$: drifts, exact $\mathcal{R}$ integrals, and Heun NFE 8 Gaussian W_2.

which holds because $2 < \pi < 8$. The four linear Heun factors are the rationals $47/52$, $6/5$, $497/370$, and $97/74$, all positive. Exact VP grid values are $a(0) = a(1) = 0$, $a(1/2) = 3\pi/10$, and $a(1/4) = (9 + 15\sqrt{2})\pi/82$, $a(3/4) = (-9 + 15\sqrt{2})\pi/82$. Positivity of $a(3/4)$ is $15^2 \cdot 2 - 9^2 = 369 > 0$. The four-step product lies in $\mathbb{Q}[\pi, \sqrt{2}]$ with nonnegative coefficients, so substituting $\sqrt{2} < 99/70$ and $\pi < 355/113$ is valid and yields $r_{\text{VP}} < 187/100$ by an integer comparison in Appendix A. Interval arithmetic is a cross-check only. □

4 Sign change and transported defects

For any independent linear coupling, $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. The VP drift vanishes at the endpoints and does not change sign on $(0, 1)$. Averaging a^2 does not retain that sign change. Fixed-stage Runge–Kutta methods use signed stage evaluations that are not determined by $\mathcal{R}$. The sign difference is consistent with the observed inversion, but the analysis does not isolate it as a causal contribution.

Let $e_n = \sqrt{q(t_{n+1})/q(t_n)}$ and let r_n be the numerical factor. The telescoping identity

$$\prod_n r_n - \prod_n e_n = \sum_n (r_n - e_n)\Big(\prod_{j<n} r_j\Big)\Big(\prod_{j>n} e_j\Big) \tag{2}$$

is an accounting tool, not the content of the inversion. The content is the leading exact-minus-method coefficient: $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-a''/12 + a^3/6)h^3$ for the implemented Heun factor. Those expansions involve different derivatives, signs, and stage locations.

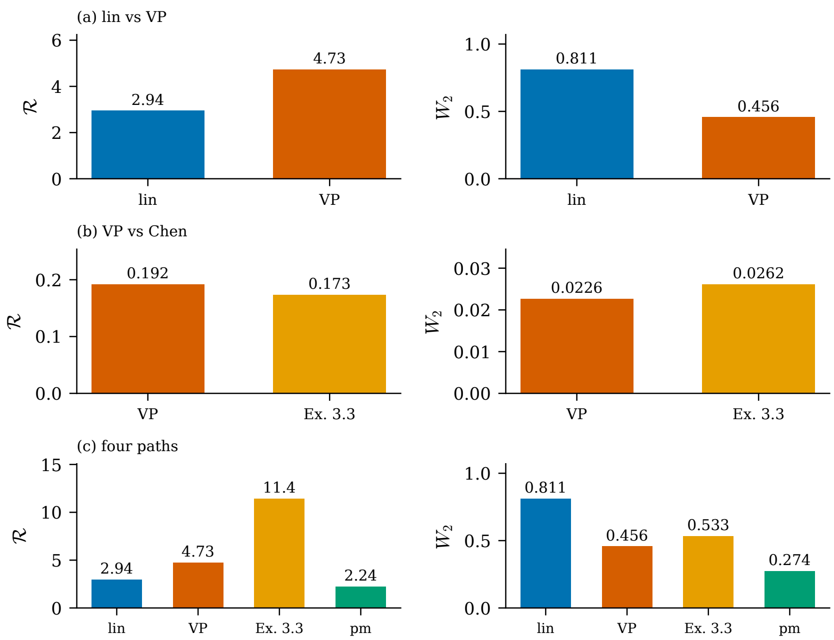

Figure 2: Three comparisons. Left: $\mathcal{R}$. Right: Euler NFE 8 W_2. (a) linear vs VP, rank-2 factor-plus-noise $d = 8$; (b) VP vs Chen Ex. 3.3, anisotropic $d = 8$; (c) four specified paths, rank-2 factor-plus-noise $d = 8$. Among those four paths, per-mode wins $\mathcal{R}$ and W_2 in 36 of 36 tested blocks.

Table 1: Four specified schedules at Heun, NFE 8. Among these four candidate paths, per-mode has the smallest W_2 on every geometry in the 36 tested blocks. When Σ_1 has at least two distinct eigenvalues, the per-mode path is not a C^1 shared interpolant.

family	d	$\mathcal{R}_{\text{lin}}$	$\mathcal{R}_{\text{VP}}$	$\mathcal{R}_{\text{sc}}$	$\mathcal{R}_{\text{log}}$	Heun-8 W_2 min
anisotropic	2	0.972	0.192	0.173	0.120	log-cov
anisotropic	8	0.972	0.192	0.173	0.120	log-cov
factor+noise	2	1.246	0.625	0.668	0.380	log-cov
rank-2 factor+noise	8	2.944	4.731	11.419	2.244	log-cov

5 Four-path enumeration and the inversion region

Source $N(0, I)$. Four centered targets: anisotropic Gaussians with condition number 4 in dimensions 2 and 8 (eigenvalues a geometric sequence from $1/2$ to 2); factor-plus-noise covariances $\Sigma = FF^\top + 0.05I$ with $F \in \mathbb{R}^{d\times 2}$ i.i.d. standard normal (seed 271828). When $d = 2$ the factor rank equals the ambient dimension. Because $\mathcal{R}$ uses the spectral norm, the two anisotropic dimensions share the same $\mathcal{R}$ pair $(0.972296, 0.192270)$, so the grid contains three distinct linear-versus-VP $\mathcal{R}$ comparisons. Each geometry is run under Euler, Heun, and RK4 at NFE $\{8, 16, 32\}$. Headline $\mathcal{R}$ is the continuous integral of $\max_i a_i(t)^2$. Ranking inversions were identified in Phase 3; the counts below are a confirmatory Phase 4 reproduction on configurations frozen before the rerun.

The per-mode log-covariance path $q_i(t) = \lambda_i^t$ is one of four specified candidate paths. Among those four paths on the stated 36 tested blocks it has the smallest $\mathcal{R}$ on every geometry and the smallest Gaussian W_2 in all 36 tested blocks (Table 1 reports Heun at NFE 8). This finite enumeration does not imply that a global $\mathcal{R}$-minimizer minimizes fixed-NFE error. When Σ_1 has at least two distinct eigenvalues, that path is not a C^1 shared interpolant. Linear versus VP invert in 5 of 12 geometry$\times$solver cells; 4 of those 12 invert at every tested NFE. As a post-hoc pairwise comparison within the shared-schedule class, trigonometric VP versus Chen Ex. 3.3 with $M = \lambda_{\max}$ invert in 9 of 36 tested blocks and 4 of 12 cells.

Table 2: Strongest centered inversion: rank-2 factor-plus-noise, $d = 8$, Euler, NFE 8. Eigenvalues: 0.05 (six times), 4.64827725629505, 10.675638226546441. Deterministic calculation: $\mathcal{R}$ is evaluated by adaptive quadrature, while W_2 uses exact Gaussian moments propagated through the modal Runge–Kutta updates.

path	$\mathcal{R}$	$\widehat{\mathcal{R}}_{24}$	Gaussian W_2
linear	$\mathbf{2.9441044083}$	2.9476523251	0.8108540111
VP	4.7305438136	4.7295206355	$\mathbf{0.4564779075}$

A one-dimensional sweep $\lambda \in [0.05, 100]$ (Figure 3) locates the linear-versus-VP inversions. At $\lambda = 4$, the inversion is present under Heun at every tested budget NFE$\in \{4, 8, 12, 16, 24, 32, 64, 128, 256\}$, absent under Euler, and mixed under RK4 (absent at NFE 8 and 12). Euler *does* invert at more extreme λ on the same sweep. The scalar example is therefore solver-dependent, not an artefact of Heun at budget 8 alone.

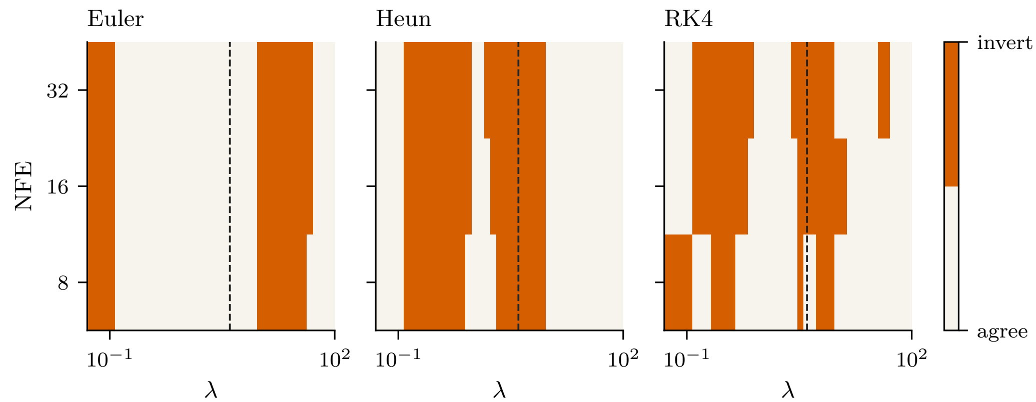

Figure 3: Linear versus VP ranking inversion versus target variance λ. Dark cells: $\mathcal{R}$ and W_2 disagree. The dashed line is $\lambda = 4$ (Proposition 1).

6 A grid-aware Class S Euler construction

Theorem 1. *Fix $L > 0$ and an integer $N \geq 1$. Let $h = 1/N$ and let forward Euler use left endpoints $t_n = nh$ on $[0, 1]$. Set $\varepsilon_N = N(e^{L/N} - 1) - L > 0$, $a_0(t) \equiv L$, and $a_{1,N}(t) = L + \varepsilon_N \cos(2\pi N t)$. For $x' = a(t)x$, $x(0) = 1$, and the Class S embeddings $q_k = \exp(2\int_0^t a_k)$, $\alpha_k = \sqrt{q_k}\cos(\pi t/2)$, $\sigma_k = \sqrt{q_k/e^{2L}}\sin(\pi t/2)$: both exact endpoints equal e^L; a_0 uniquely minimizes $\int a^2$ among scalar fields with integral L; Euler is exact for $a_{1,N}$ and strictly inexact for a_0. Endpoint error is $|r - e^L|$, equivalently Gaussian W_2 between $N(0, r^2)$ and $N(0, e^{2L})$.*

The competitor is constructed as a function of the solver grid. Phase offsets and non-grid frequencies destroy exact Euler aliasing. This is a solver-grid-dependent impossibility result, not a model of natural schedules, and not the mechanism of Proposition 1.

7 Related work

Flow matching and interpolants supply the Gaussian family [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023, Tong et al., 2024]. Lipschitz-guided design proposes the selection criterion tested [Chen et al., 2025]. Solver-specific schedules and exact linear parts are documented in diffusion practice [Sabour et al., 2024, Karras et al., 2022, Lu et al., 2022, Zhang and Chen, 2023]. Stéphanovitch [2026] prove sharp Lipschitz regularity of canonical flow-matching fields and Euler-type W_2 rates; that is a bound, not a pairwise ranking of two interpolants at equal NFE.

Tao and Choi [2026] decompose $\nabla v = S + \Omega$ and control Euler error by the logarithmic norm $\mu_2 = \lambda_{\max}(S)$, which is signed, whereas $\mathcal{R}$ integrates $\|A\|_2^2$. In one dimension $\mu_2(a) = a$ and $\|A\|_2 = |a|$. Their Theorem 5 states that McCann displacement interpolation has vanishing material derivative, hence Euler is order 2 in the Eulerian analysis; Remark 7 records the stronger Lagrangian exactness observed on that interpolant. Neither statement applies to the independent-coupling linear or VP path studied here. The present paper supplies a rationally certified Gaussian instance of ranking failure for the unsigned average $\|A\|_2^2$. A counterexample to μ_2 itself is outside our scope. Benton et al. [2024] bound W_2 of a learned flow; that is an error bound, not a pairwise ranking. Consistency models [Song et al., 2023] distill few-step maps.

8 Limitations

The systems are commuting affine Gaussian interpolants, three fixed-step Runge–Kutta methods, and small equal-NFE budgets. No learned neural field is evaluated. Chen et al. do not claim or prove a discretization-error bound, nor that A_2 universally ranks equal-NFE solver error. Among four specified candidate paths, the per-mode path wins 36 of 36 tested blocks; that does not imply that a global $\mathcal{R}$-minimizer minimizes fixed-NFE error. Theorem 1 is grid-aligned by construction. Cell counts are hierarchical descriptive summaries, not population rates.

9 Conclusion

An integrated unsigned Jacobian-energy functional does not contain enough solver-specific temporal information to universally determine few-step endpoint error. Averaged squared Jacobian regularity can misrank equal-NFE Gaussian W_2 already for two C^1 interpolants of independent $N(0,1)$ and $N(0,4)$. A grid-aware construction shows that even the global $\mathcal{R}$-minimizer need not minimize fixed-grid Euler error. A specified finite enumeration contains both agreement (36 of 36 tested blocks among four candidate paths) and pairwise disagreement.

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A Rational VP Heun certificate

This appendix records the integer comparison used in Proposition 1. It is part of this PDF, not a separate supplement file.

The four-step VP Heun product is the following element $P(\pi, \sqrt{2})$ of $\mathbb{Q}[\pi, \sqrt{2}]$, obtained by expanding the four Heun factors with $h = 1/4$ and reducing $\sqrt{2}^2 = 2$:

$$\begin{aligned} P(\pi, \sqrt{2}) = 1 &+ \tfrac{3}{40}\pi + \tfrac{15}{164}\pi\sqrt{2} + \tfrac{78579}{10758400}\pi^2 + \tfrac{45}{5248}\pi^2\sqrt{2} + \tfrac{11421}{17213440}\pi^3 + \tfrac{4671}{17213440}\pi^3\sqrt{2} \\ &+ \tfrac{5103}{275415040}\pi^4 + \tfrac{567}{55083008}\pi^4\sqrt{2} + \tfrac{243}{2203320320}\pi^5 + \tfrac{729}{2203320320}\pi^5\sqrt{2} + \tfrac{729}{176265625600}\pi^6. \end{aligned} \tag{3}$$

Each of the twelve coefficients is a ratio of nonnegative integers, hence nonnegative, and the constant term is 1. Substituting $\pi < 355/113$ and $\sqrt{2} < 99/70$ is therefore valid and yields

$$P\Big(\frac{355}{113}, \frac{99}{70}\Big) = \frac{19211335367141247013930 3}{10275342787993970881331 2} < \frac{187}{100},$$

equivalently

$$1921133536714124701393030 0 < 1921489101354872554808934 4.$$

Hence $r_{\mathrm{VP}} < 187/100$ and $W_2^{\mathrm{VP}} > 13/100$. The bound $\sqrt{2} < 99/70$ is $99^2 - 2 \cdot 70^2 = 1$; $\pi < 355/113$ follows from Machin's formula with Leibniz remainders (seven Taylor terms for $\arctan(1/5)$, four for $\arctan(1/239)$). An independent `mpmath` 1.3.0 interval enclosure is a cross-check, not the proof.